

Viewing Files & Logs Cheat Sheet

Read files, jump to the part you need, and watch logs live: cat, less, head, tail and watch.

Which one to use

Command	Use it when	Command	Use it when
cat file	The file is short; dump it all	tail file	You want the end (last 10 lines)
less file	The file is long; scroll and search	tail -F file	You want to watch new lines arrive
head file	You want the start (first 10 lines)	watch cmd	You want to rerun a command every 2 s

head and tail

Option	What it does	Option	What it does
head -n 20	First 20 lines	tail -f	Follow: print new lines as they're added
tail -n 50	Last 50 lines	tail -F	Follow, and keep going after the log rotates
tail -n +2	From line 2 to the end (skip a header)	tail -f a b	Follow several files at once
head -n -5	All but the last 5 lines (GNU)	--pid=1234	Stop when process 1234 exits (GNU)

Press **Ctrl-C** to stop following. **-f** goes quiet after log rotation; **-F** re-opens it by name.

less: moving and searching

Key	Does	Key	Does
Space / b	Page down / page up	/word	Search forward (?word = backward)
g / G	Jump to the start / end	&word	Show only lines containing word
F	Follow like tail -f ; Ctrl-C to stop	j / k	One line down / up (arrows work too)
q	Quit	-N -S -i -R	Line numbers · no wrap · any case · colors

Reading tail output

```
$ tail -F /var/log/nginx/access.log /var/log/nginx/error.log
==> /var/log/nginx/access.log <== ← header: the lines below come from this file
203.0.113.7 - - [03/Oct/2026:14:02:11 +0000] "GET / HTTP/1.1" 200 612
==> /var/log/nginx/error.log <== ← output switches files as new lines arrive
2026/10/03 14:02:15 [error] 812#812: *41 open() failed (2: No such file)
tail: '/var/log/nginx/access.log' has been replaced; following new file ← -F at work
```

Where logs live

File	Contains	File	Contains
/var/log/syslog	Main system log (Debian, Ubuntu)	/var/log/messages	Main system log (RHEL, Fedora)
/var/log/auth.log	Logins and sudo (Debian, Ubuntu)	/var/log/secure	Logins and sudo (RHEL, Fedora)
/var/log/nginx/	Web server access and error logs	/var/log/kern.log	Kernel messages (also dmesg)
syslog.1 .2.gz	Older, rotated copies	journalctl	systemd's log; may be the only log

Handy recipes

Command	Does
tail -F /var/log/syslog grep --line-buffered error	Watches only new error lines, live
less +F /var/log/syslog	Follows live; Ctrl-C to scroll back, F to resume
zless /var/log/syslog.2.gz	Reads a compressed log (zcat , zgrep also work)
zgrep 'Failed password' /var/log/auth.log*	Searches current and rotated logs together
cat -A file	Shows tabs as ^I and line ends as \$ (GNU)
watch -n 5 -d 'df -h'	Reruns every 5 s and highlights what changed

Remember: use **tail -F** (capital F) for logs so it keeps working after log rotation. Piping a live log into grep? Add **--line-buffered** or matches show up late.